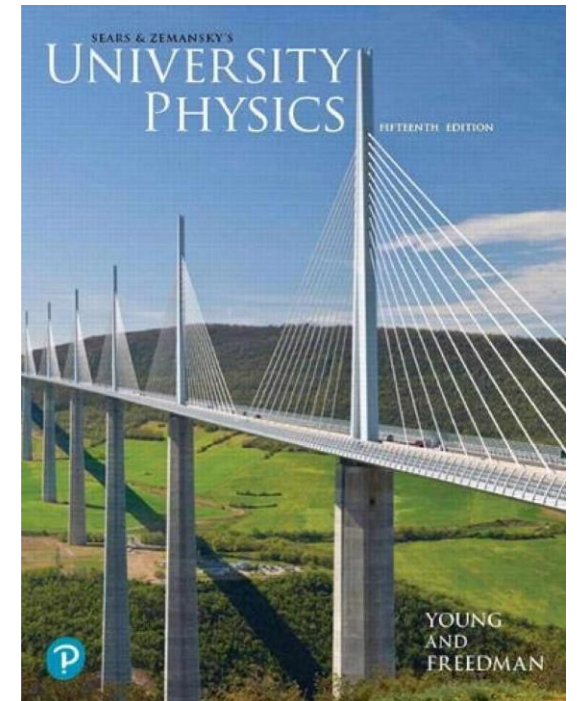


普通物理I PHYS1181.03

平衡和弹性

Equilibrium and Elasticity



本节课:

A. 平衡条件

B. 应变、应力和胡克定律

a) 拉应变和压应变

b) 体应变

c) 切应变

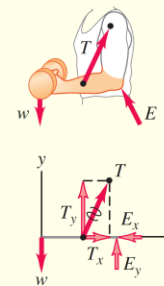
d) 胡克定律的局限性

Conditions for equilibrium: For a rigid body to be in equilibrium, two conditions must be satisfied. First, the vector sum of forces must be zero. Second, the sum of torques about any point must be zero. The torque due to the weight of an object can be found by assuming the entire weight is concentrated at the center of gravity, which is at the same point as the center of mass if $\vec{g}$ has the same value at all points. (See Examples 11.1–11.4.)

$$\sum \vec{F} = \mathbf{0} \quad (11.1)$$

$$\sum \vec{\tau} = \mathbf{0} \quad \text{about any point} \quad (11.2)$$

$$\vec{r}_{\text{cm}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \cdots}{m_1 + m_2 + m_3 + \cdots} \quad (11.4)$$

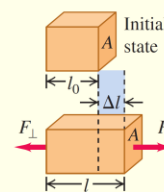


Stress, strain, and Hooke's law: Hooke's law states that in elastic deformations, stress (force per unit area) is proportional to strain (fractional deformation). The proportionality constant is called the elastic modulus.

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic modulus} \quad (11.7)$$

Tensile and compressive stress: Tensile stress is tensile force per unit area, $F_{\perp}/A$. Tensile strain is fractional change in length, $\Delta l/l_0$. The elastic modulus for tension is called Young's modulus Y . Compressive stress and strain are defined in the same way. (See Example 11.5.)

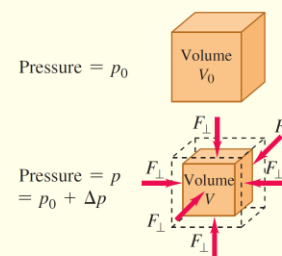
$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} = \frac{F_{\perp}/A}{\Delta l/l_0} = \frac{F_{\perp}}{A} \frac{l_0}{\Delta l} \quad (11.10)$$



Bulk stress: Pressure in a fluid is force per unit area. Bulk stress is pressure change, Δp , and bulk strain is fractional volume change, $\Delta V/V_0$. The elastic modulus for compression is called the bulk modulus, B . Compressibility, k , is the reciprocal of bulk modulus: $k = 1/B$. (See Example 11.6.)

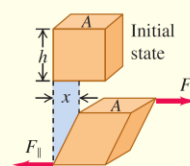
$$p = \frac{F_{\perp}}{A} \quad (11.11)$$

$$B = \frac{\text{Bulk stress}}{\text{Bulk strain}} = -\frac{\Delta p}{\Delta V/V_0} \quad (11.13)$$



Shear stress: Shear stress is force per unit area, $F_{\parallel}/A$, for a force applied tangent to a surface. Shear strain is the displacement x of one side divided by the transverse dimension h . The elastic modulus for shear is called the shear modulus, S . (See Example 11.7.)

$$S = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F_{\parallel}/A}{x/h} = \frac{F_{\parallel}}{A} \frac{h}{x} \quad (11.17)$$



The limits of Hooke's law: The proportional limit is the maximum stress for which stress and strain are proportional. Beyond the proportional limit, Hooke's law is not valid. The elastic limit is the stress beyond which irreversible deformation occurs. The breaking stress, or ultimate strength, is the stress at which the material breaks.

Examples

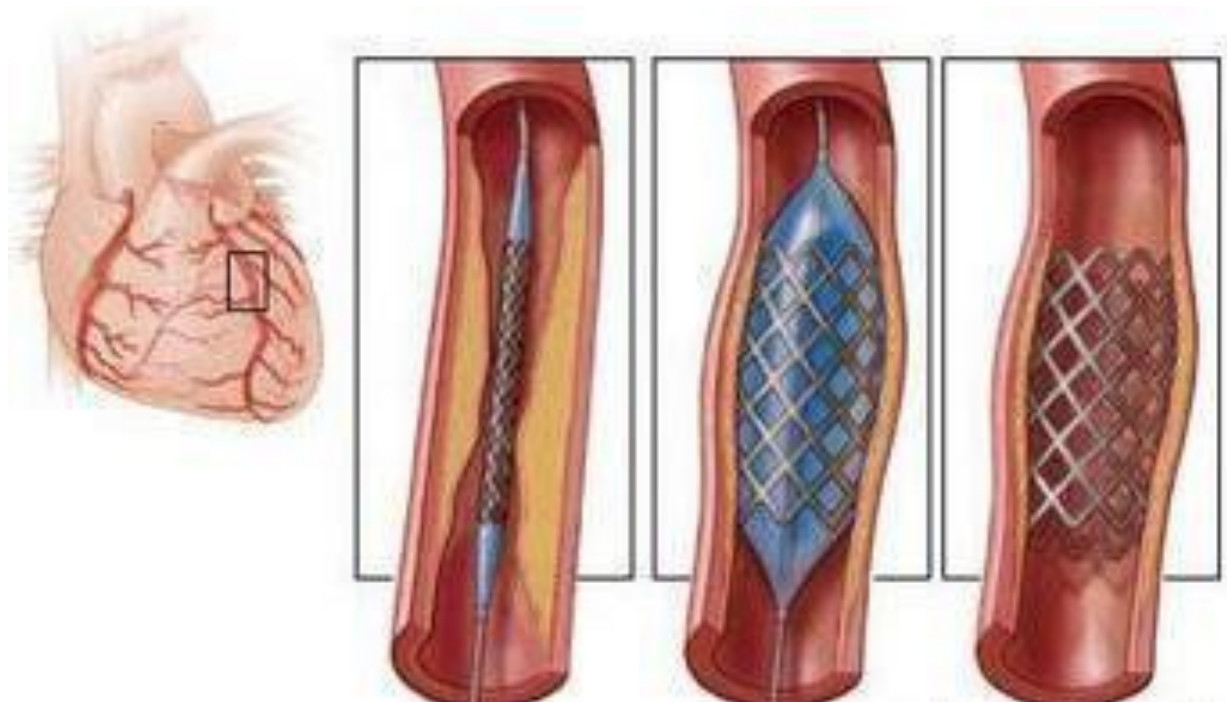


Examples

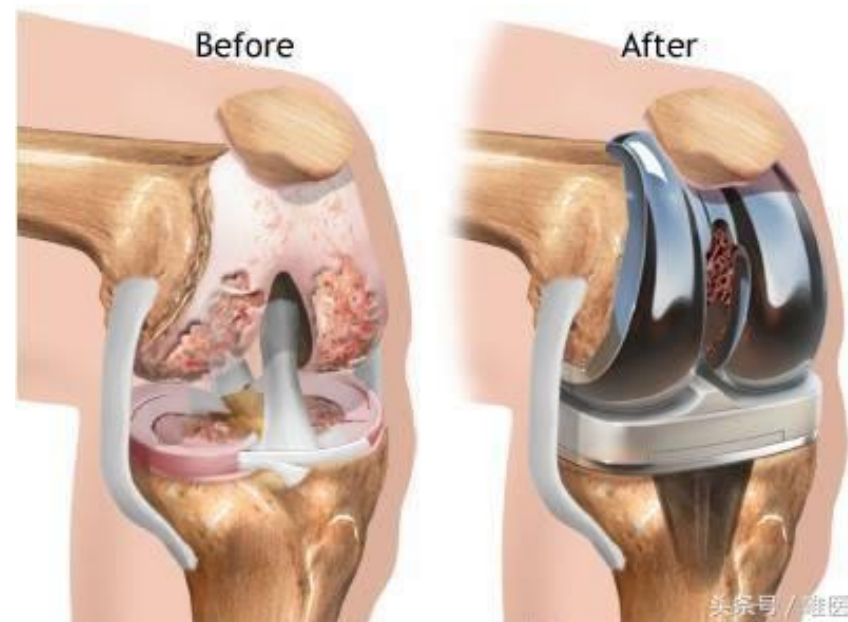


Examples

心脏支架



人工膝盖



A. 平衡条件

什么是平衡？

➤ 没有加速度



直线加速度
 $=0$

&

转动加速度
 $=0$



力为零



力矩为零

平衡的两个条件

First condition for equilibrium:

For the center of mass of an object at rest to remain at rest ...

$$\sum \vec{F} = \mathbf{0}$$

... the *net external force* on the object must be *zero*.

Second condition for equilibrium:

For a nonrotating object to remain nonrotating ...

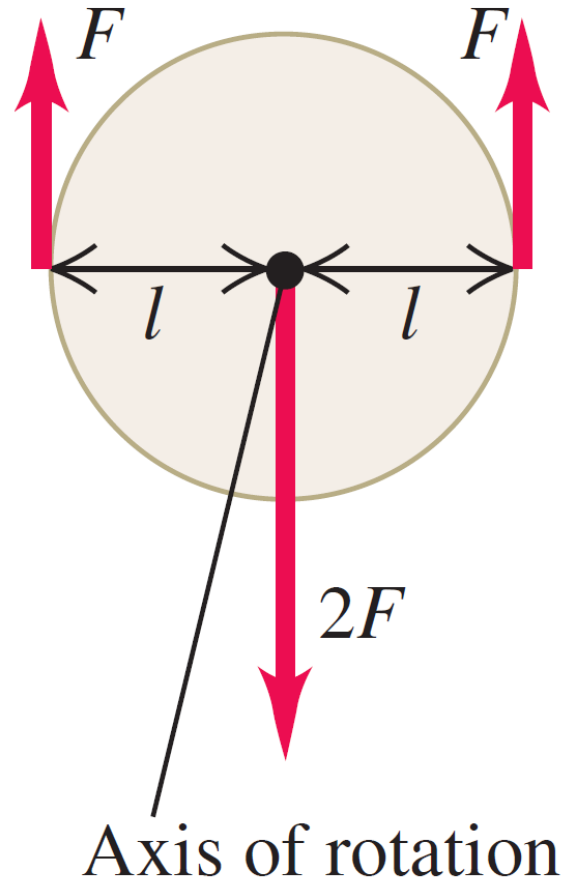
$$\sum \vec{\tau} = \mathbf{0}$$

...the *net external torque* around any point on the object must be *zero*.

质点的平衡 -- 牛顿第一定律: **合力为零**

刚体的平衡 -- **合力为零 + 合力矩为零**

Example 1



First condition satisfied:

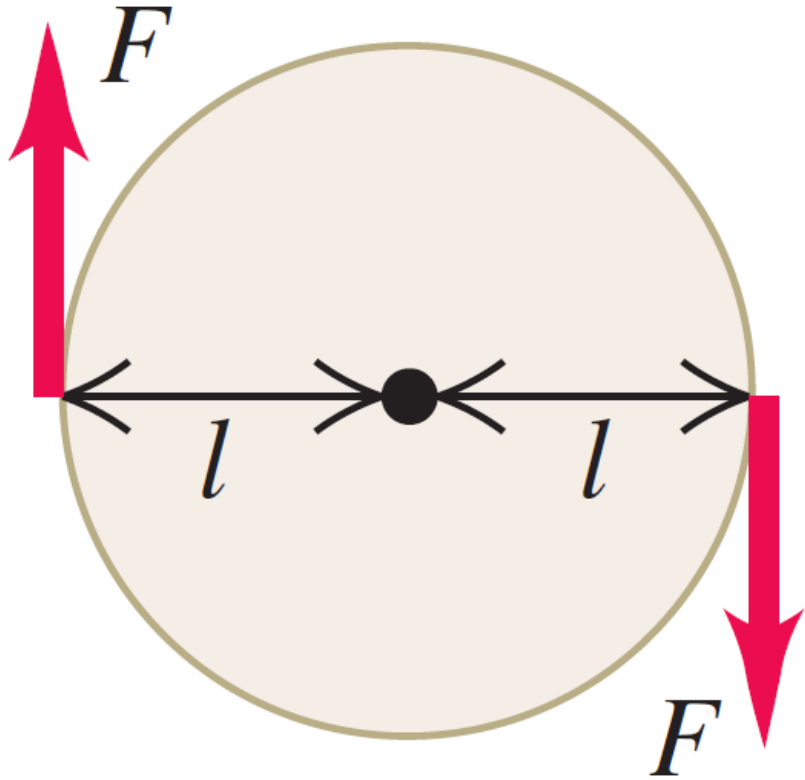
Net force $= 0$, so object at rest has no tendency to start moving as a whole.

Second condition satisfied:

Net torque about the axis $= 0$, so object at rest has no tendency to start rotating.

✓ In static equilibrium (静态平衡)

Example 2



First condition satisfied:

Net force = 0, so object at rest has no tendency to start moving as a whole.

Second condition NOT

satisfied: There is a net clockwise torque about the axis, so object at rest will start rotating clockwise.

x Not in equilibrium (非平衡)

Real examples of dynamic equilibrium



质心和重心

center of mass and center of gravity

质心定义:

Position vector of center of mass of a system of particles $\vec{r}_{\text{cm}}$

Position vectors of individual particles

Masses of individual particles

$$\vec{r}_{\text{cm}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$$

什么时候质心和重心重合?

---重力加速度在整个质心系中不变时，二者重合。

质点系所受重力力矩

单个质点所受重力力矩:

$$\vec{\tau}_i = \vec{r}_i \times \vec{w}_i = \vec{r}_i \times m_i \vec{g}$$

所有质点所受重力力矩:

$$\begin{aligned}\vec{\tau} &= \sum_i \vec{\tau}_i = \vec{r}_1 \times m_1 \vec{g} + \vec{r}_2 \times m_2 \vec{g} + \cdots \\ &= (m_1 \vec{r}_1 + m_2 \vec{r}_2 + \cdots) \times \vec{g} \\ &= \left(\sum_i m_i \vec{r}_i \right) \times \vec{g}\end{aligned}$$

质心所受重力力矩

所有质点所受重力力矩:

$$\begin{aligned}\vec{\tau} &= \sum_i \vec{\tau}_i = \vec{r}_1 \times m_1 \vec{g} + \vec{r}_2 \times m_2 \vec{g} + \cdots \\ &= (m_1 \vec{r}_1 + m_2 \vec{r}_2 + \cdots) \times \vec{g} \\ &= \left(\sum_i m_i \vec{r}_i \right) \times \vec{g}\end{aligned}$$

质心所受重力力矩:

$$\vec{\tau} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \cdots}{m_1 + m_2 + \cdots} \times M \vec{g} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} \times M \vec{g}$$

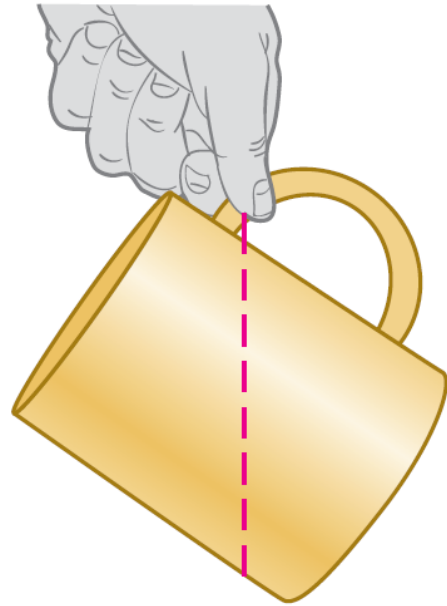


$$\vec{\tau} = \vec{r}_{\text{cm}} \times M \vec{g} = \vec{r}_{\text{cm}} \times \vec{w}$$

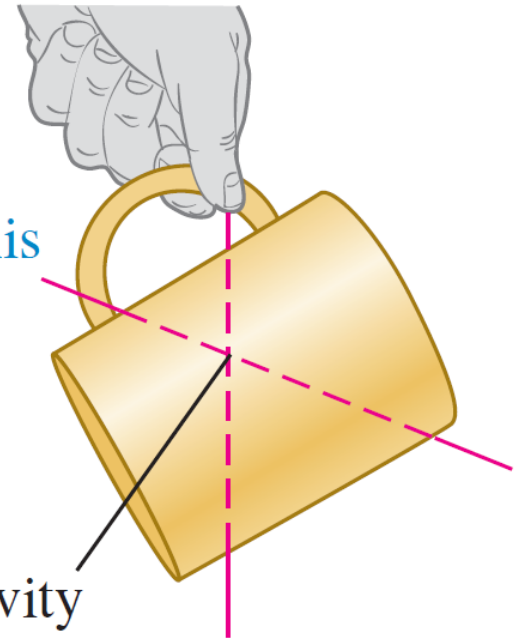
怎样找不规则物体的重心？

Where is the center of gravity of this mug?

① Suspend the mug from any point. A vertical line extending down from the point of suspension passes through the center of gravity.

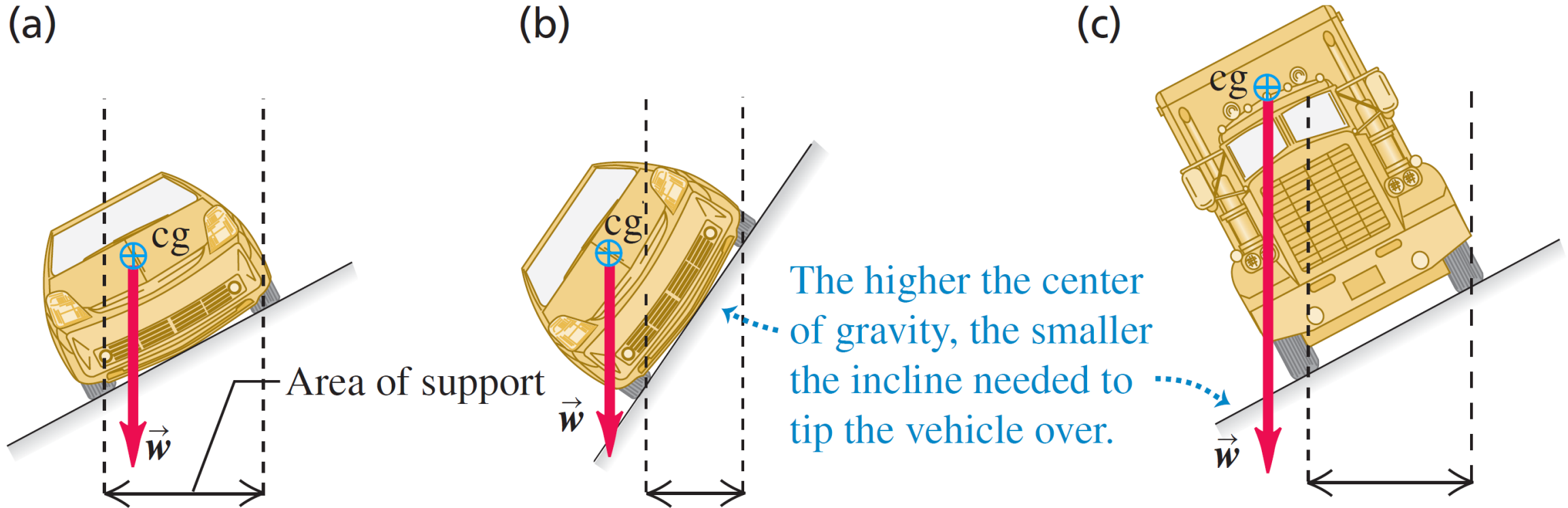


② Now suspend the mug from a different point. A vertical line extending down from this point intersects the first line at the center of gravity (which is inside the mug).



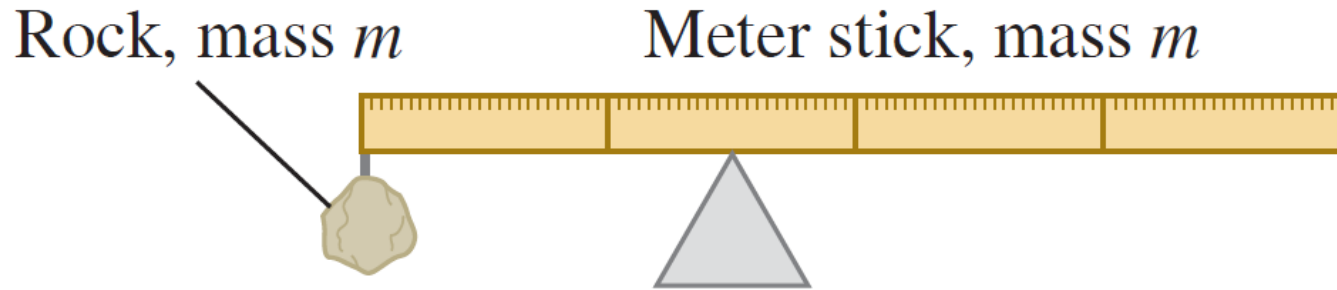
Center of gravity

思考：下列那些情况不能保持平衡？



问题:

Figure 11.7 At what point will the meter stick with rock attached be in balance?



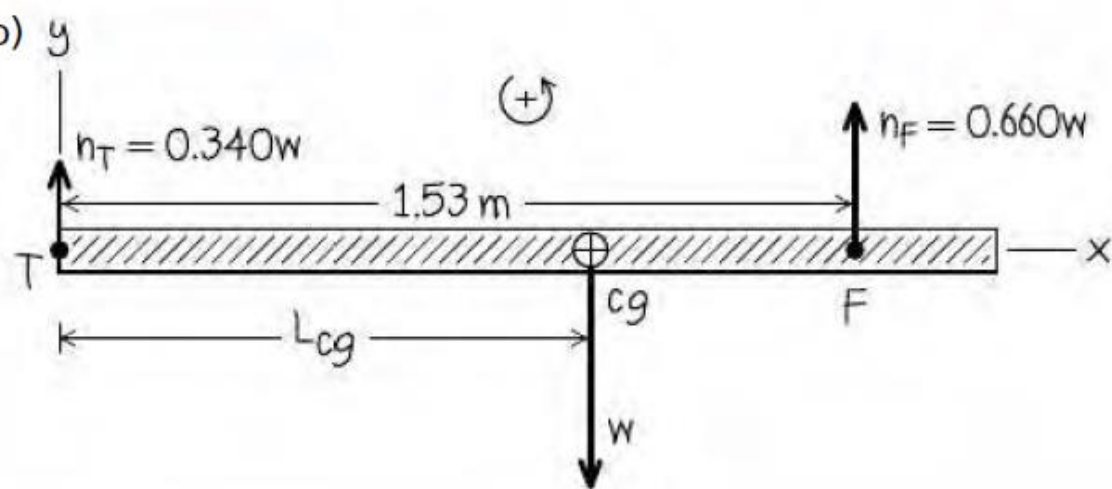
- A. Less than 0.25 m
- B. 0.25 m
- C. Between 0.25 m and 0.5 m
- D. 0.5 m
- E. More than 0.5 m

例题:

(a)



(b)



B. 应变、应力和胡克定律

1. 什么是应变？什么是应力？



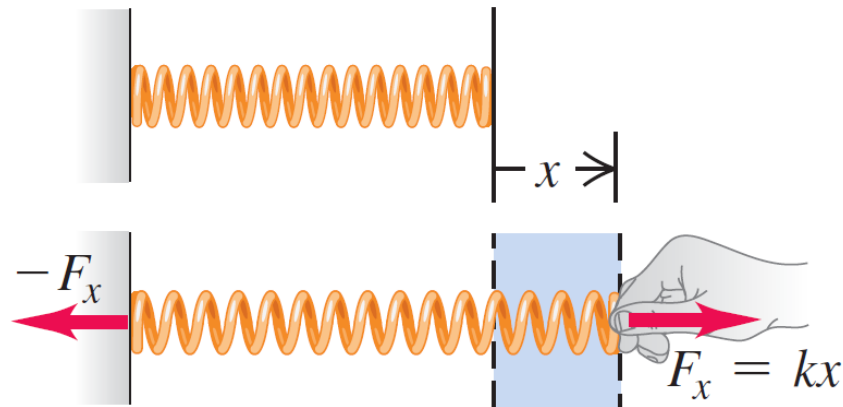
应变：物体受力后的
形变；

$$\varepsilon_x = \frac{\Delta x}{x}$$

应力：物体单位面积
上受力的大小。

$$\sigma_x = \frac{F_x}{A}$$

胡克定律 Hook's Law



Measure of forces applied to deform an object

Hooke's law: $\frac{\text{Stress}}{\text{Strain}} = \text{Elastic modulus}$ Property of material of which object is made

Measure of how much deformation results from stress

胡克定律说明：应力线性正比于应变

生活中的应力/应变例子

压应力/应变
Compressive
stress/strain



拉应力/应变
Tensile
stress/strain



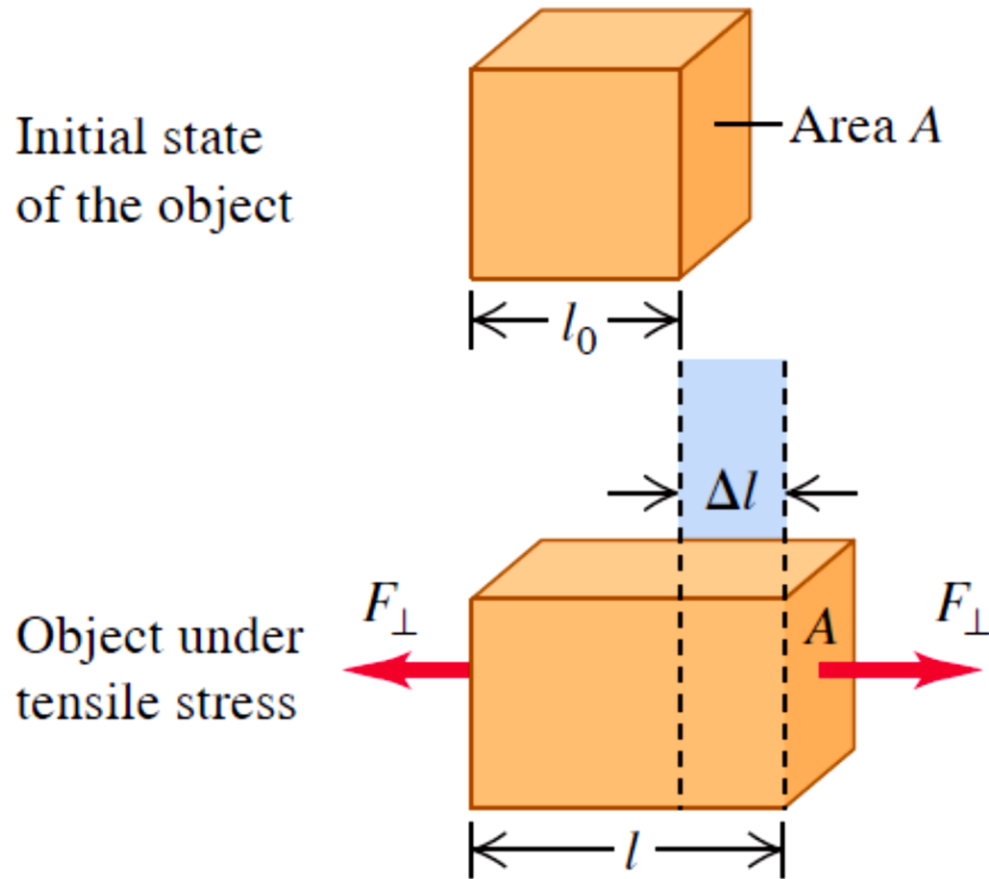
体应力/应变
Bulk
stress/strain



剪应力/应变
Shear
stress/strain



a) 拉应力/应变 Tensile stress/strain



$$\text{Tensile stress} = \frac{F_{\perp}}{A}$$

$$\text{Tensile strain} = \frac{l - l_0}{l_0} = \frac{\Delta l}{l_0}$$

应力 (stress) 单位: Pa in SI; psi in British

应变 (strain) 单位: none

1 psi = 6895 Pa

弹性形变的杨氏模量Y

Young's modulus
for tension

Force applied perpendicular
to cross section

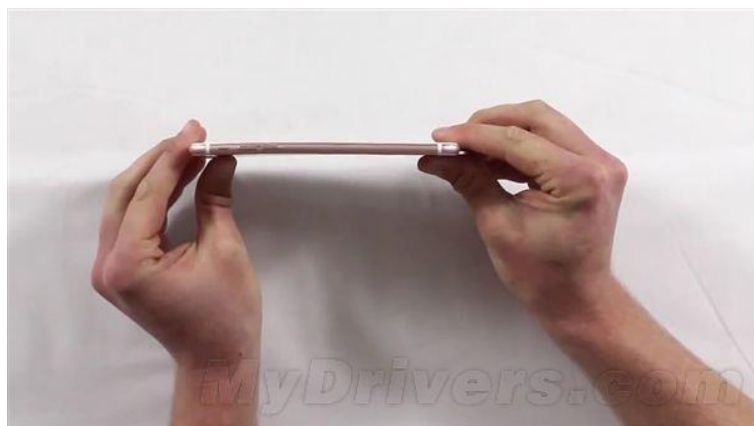
$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} = \frac{F_{\perp}/A}{\Delta l/l_0} = \frac{F_{\perp}}{A} \frac{l_0}{\Delta l}$$

Original length
(see Fig. 11.14)

Elongation
(see Fig. 11.14)

Cross-sectional area of object

适用



不适用



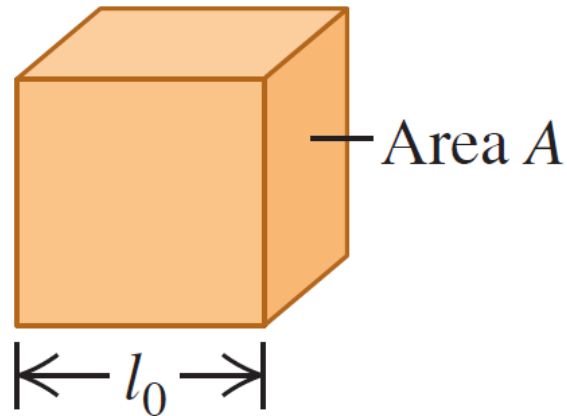
常见材料的杨氏模量

TABLE 11.1 Approximate Elastic Moduli

Material	Young's Modulus, <i>Y</i> (Pa)	Bulk Modulus, <i>B</i> (Pa)	Shear Modulus, <i>S</i> (Pa)
Aluminum	7.0×10^{10}	7.5×10^{10}	2.5×10^{10}
Brass	9.0×10^{10}	6.0×10^{10}	3.5×10^{10}
Copper	11×10^{10}	14×10^{10}	4.4×10^{10}
Iron	21×10^{10}	16×10^{10}	7.7×10^{10}
Lead	1.6×10^{10}	4.1×10^{10}	0.6×10^{10}
Nickel	21×10^{10}	17×10^{10}	7.8×10^{10}
Silicone rubber	0.001×10^{10}	0.2×10^{10}	0.0002×10^{10}
Steel	20×10^{10}	16×10^{10}	7.5×10^{10}
Tendon (typical)	0.12×10^{10}	—	—

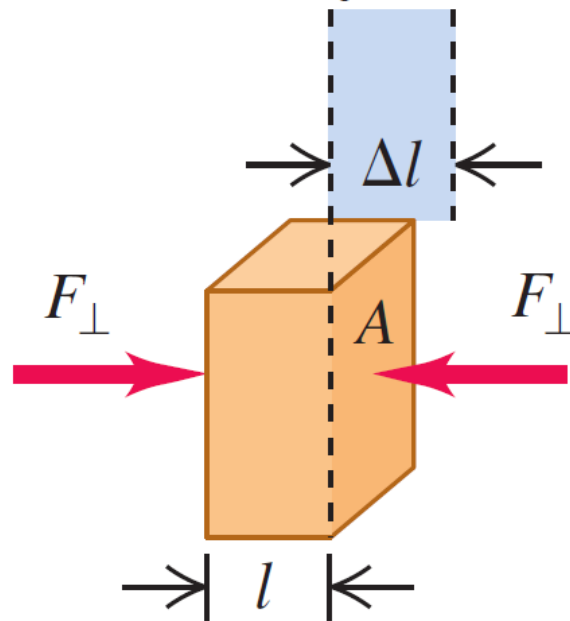
压应力/应变 Compressive stress/strain

Initial state
of the object



$$\text{Compressive stress} = \frac{F_{\perp}}{A}$$

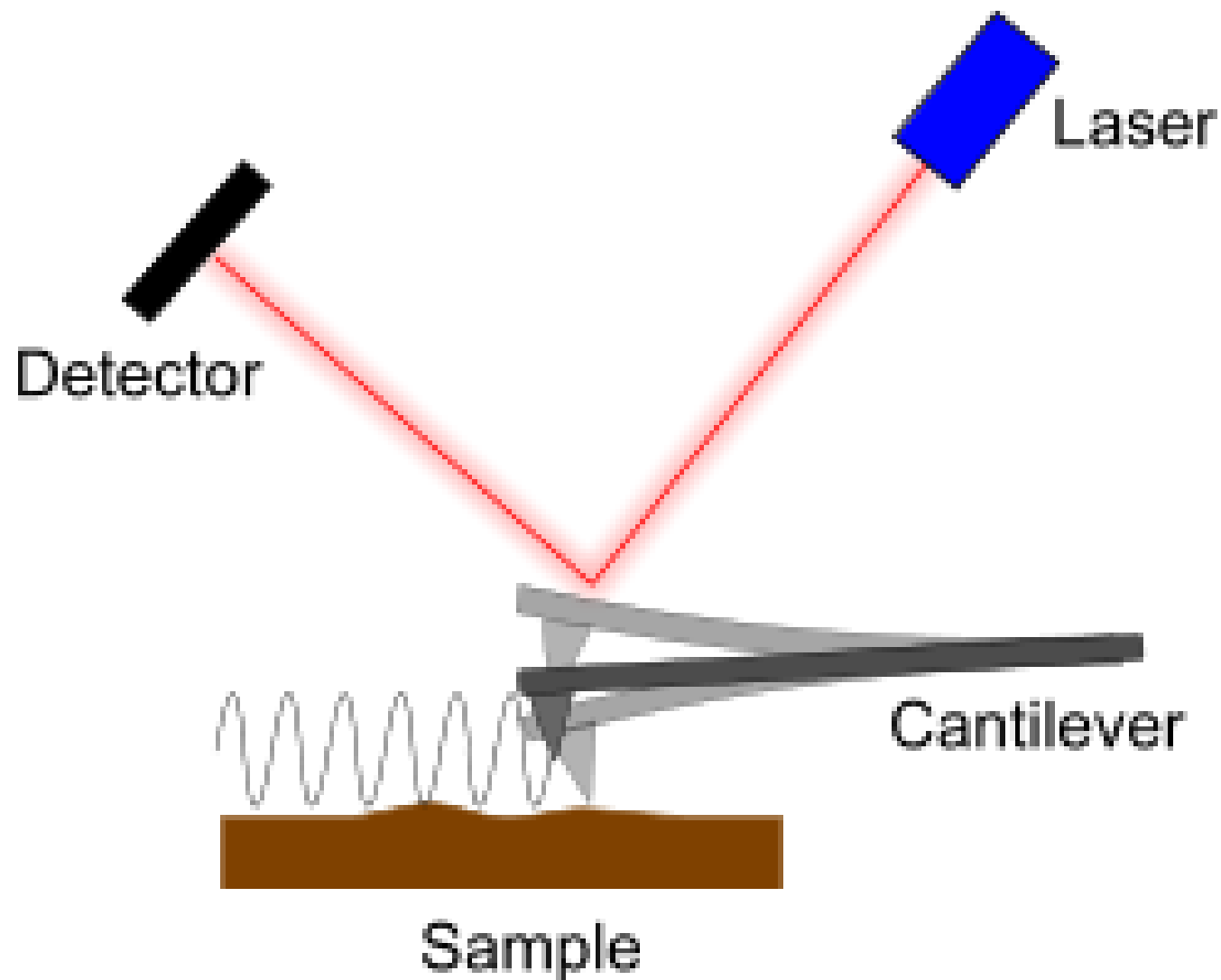
Object under
compressive
stress



$$\text{Compressive strain} = \frac{\Delta l}{l_0}$$

与拉应力/应变的区别：符号相反

例子：原子力显微镜-激光反射探测微小应变



例题

一根长2.0米的钢棒的横截面积为0.3平方厘米。它的一端挂在支架上，另一端挂着一个550公斤重的磨床。钢的杨氏模量为 20×10^{10} ，确定在杆上的应力和由此产生的应变和伸长率。

$$\text{Tensile stress} = \frac{F_{\perp}}{A} = \frac{(550 \text{ kg})(9.8 \text{ m/s}^2)}{3.0 \times 10^{-5} \text{ m}^2} = 1.8 \times 10^8 \text{ Pa}$$

$$\text{Strain} = \frac{\Delta l}{l_0} = \frac{\text{Stress}}{Y} = \frac{1.8 \times 10^8 \text{ Pa}}{20 \times 10^{10} \text{ Pa}} = 9.0 \times 10^{-4}$$

$$\begin{aligned} \text{Elongation} &= \Delta l = (\text{Strain}) \times l_0 \\ &= (9.0 \times 10^{-4})(2.0 \text{ m}) = 0.0018 \text{ m} = 1.8 \text{ mm} \end{aligned}$$

b) 体应力/应变 Bulk stress/strain



深海的潜水员和鱼

体应力/应变 (Bulk stress/strain) 的定义:

物体所受各向同性 (isotropic) 的应力和应变

物体在流体中所受各向同性压强:

Pressure in a fluid $\rightarrow p = \frac{F_{\perp}}{A}$

Force that fluid applies to surface of an immersed object

Area over which force is exerted

体应变: Bulk (volume) strain = $\frac{\Delta V}{V_0}$

体弹性模量 (Bulk modulus) :

$$\text{Bulk modulus for compression} \rightarrow B = \frac{\text{Bulk stress}}{\text{Bulk strain}} = - \frac{\Delta p}{\Delta V/V_0}$$

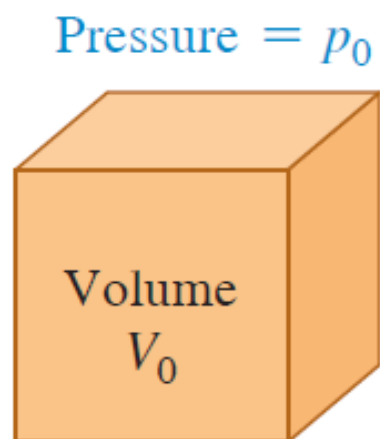
Change in volume (see Fig. 11.17) $\rightarrow$ Additional pressure on object $\rightarrow$ Original volume (see Fig. 11.17)

压缩系数 (compressibility) :

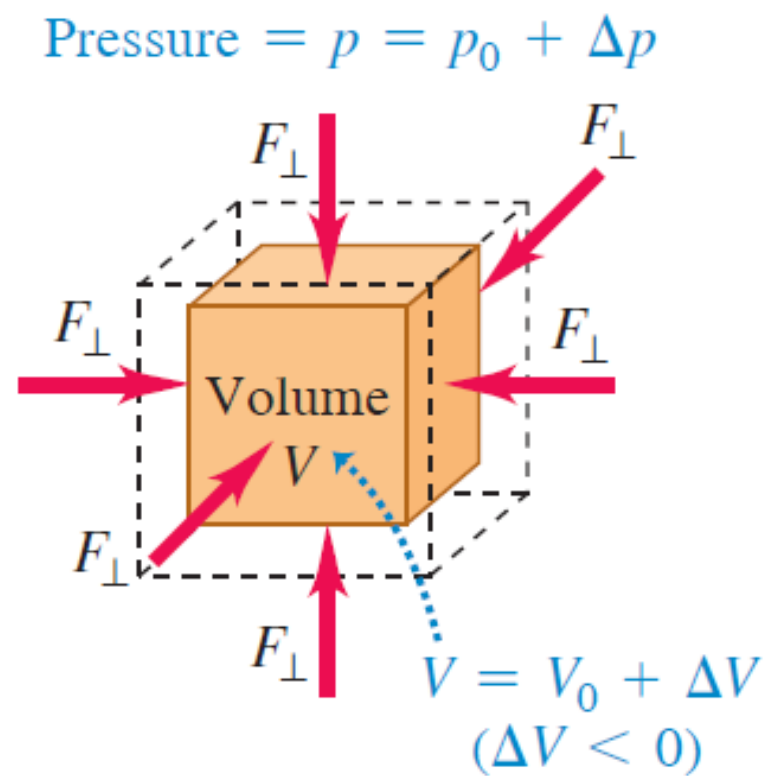
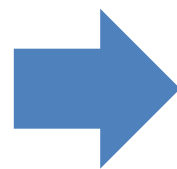
$$k = \frac{1}{B} = - \frac{\Delta V/V_0}{\Delta p} = - \frac{1}{V_0} \frac{\Delta V}{\Delta p}$$

图示：

Initial state
of the object



Object under
bulk stress



Bulk stress = Δp

Bulk strain = $\frac{\Delta V}{V_0}$

例题:

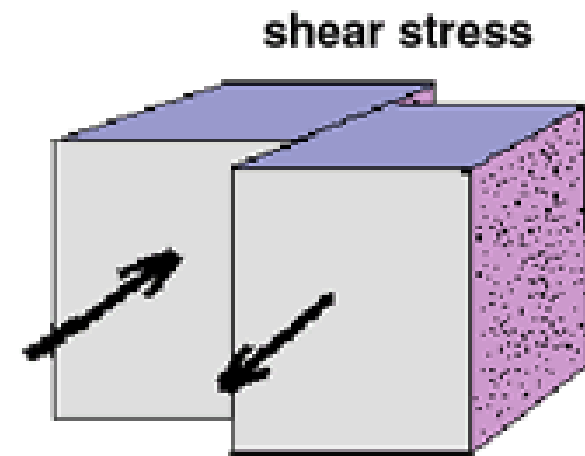
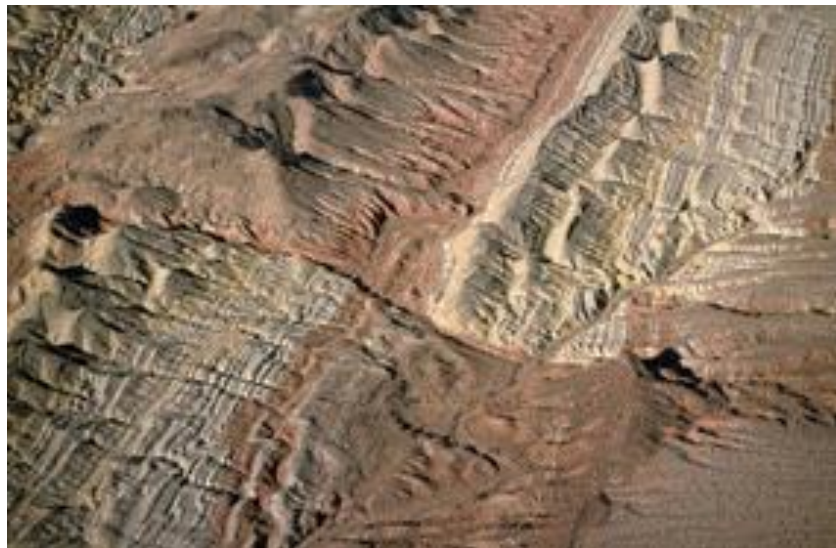
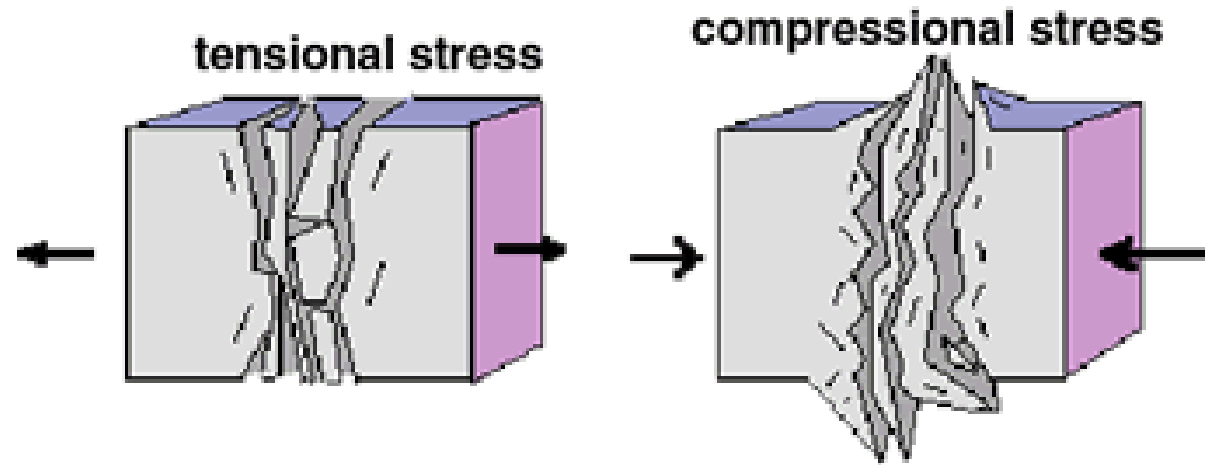
一台液压机可装 0.25m^3 (250升)的油。当压力增加 $\Delta p = 1.6 \times 10^7\text{Pa}$ (约160 atm或2300 psi)时, 计算油体积的减少量。油的体积模量为 $B = 5.0 \times 10^9\text{Pa}$ (约 $5.0 \times 10^4\text{atm}$), 可压缩性为 $k = 1/B = 20 \times 10^{-6}\text{atm}^{-1}$

$$\begin{aligned}\Delta V &= -\frac{V_0 \Delta p}{B} = -\frac{(0.25\text{ m}^3)(1.6 \times 10^7\text{ Pa})}{5.0 \times 10^9\text{ Pa}} \\ &= -8.0 \times 10^{-4}\text{ m}^3 = -0.80\text{ L}\end{aligned}$$

或者:
$$\begin{aligned}\Delta V &= -kV_0\Delta p = -(20 \times 10^{-6}\text{ atm}^{-1})(0.25\text{ m}^3)(160\text{ atm}) \\ &= -8.0 \times 10^{-4}\text{ m}^3\end{aligned}$$

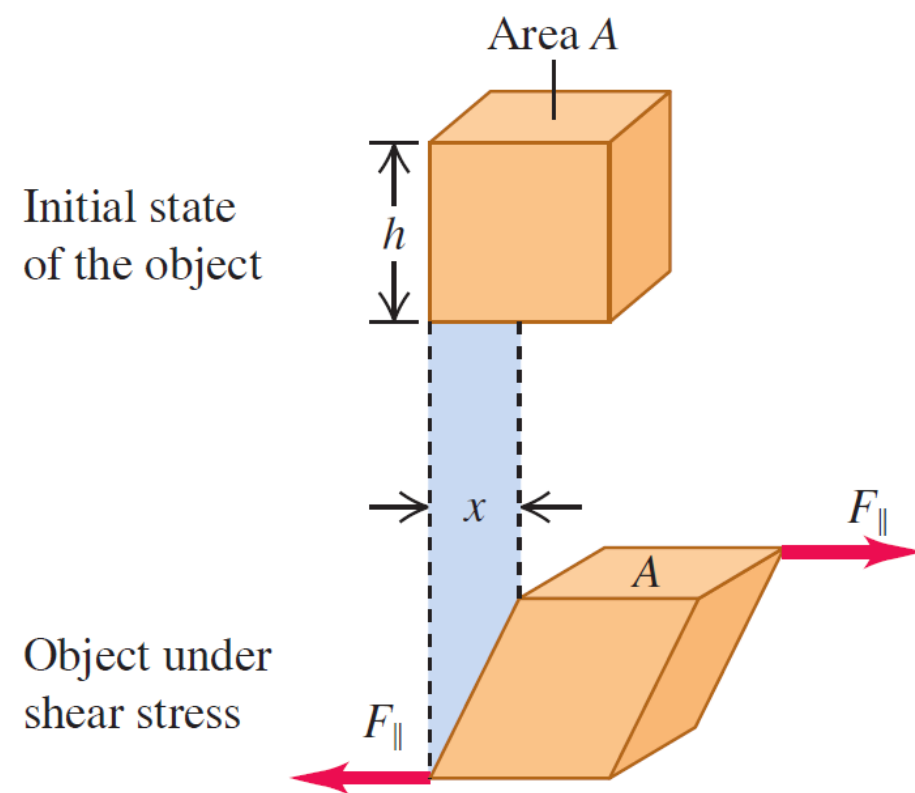
$$\frac{\Delta V}{V_0} = \frac{-8.0 \times 10^{-4}\text{ m}^3}{0.25\text{ m}^3} = -0.0032 \quad \text{or} \quad -0.32\%$$

c) 切应力/应变 Shear stress/strain



切应力/应变 Shear stress/strain定义:

Forces of equal magnitude but opposite direction act tangent to the surfaces of opposite ends of the object.



切应力:

$$\text{Shear stress} = \frac{F_{\parallel}}{A}$$

切应变:

$$\text{Shear strain} = \frac{x}{h}$$

切向模量:

Force applied tangent to surface of object

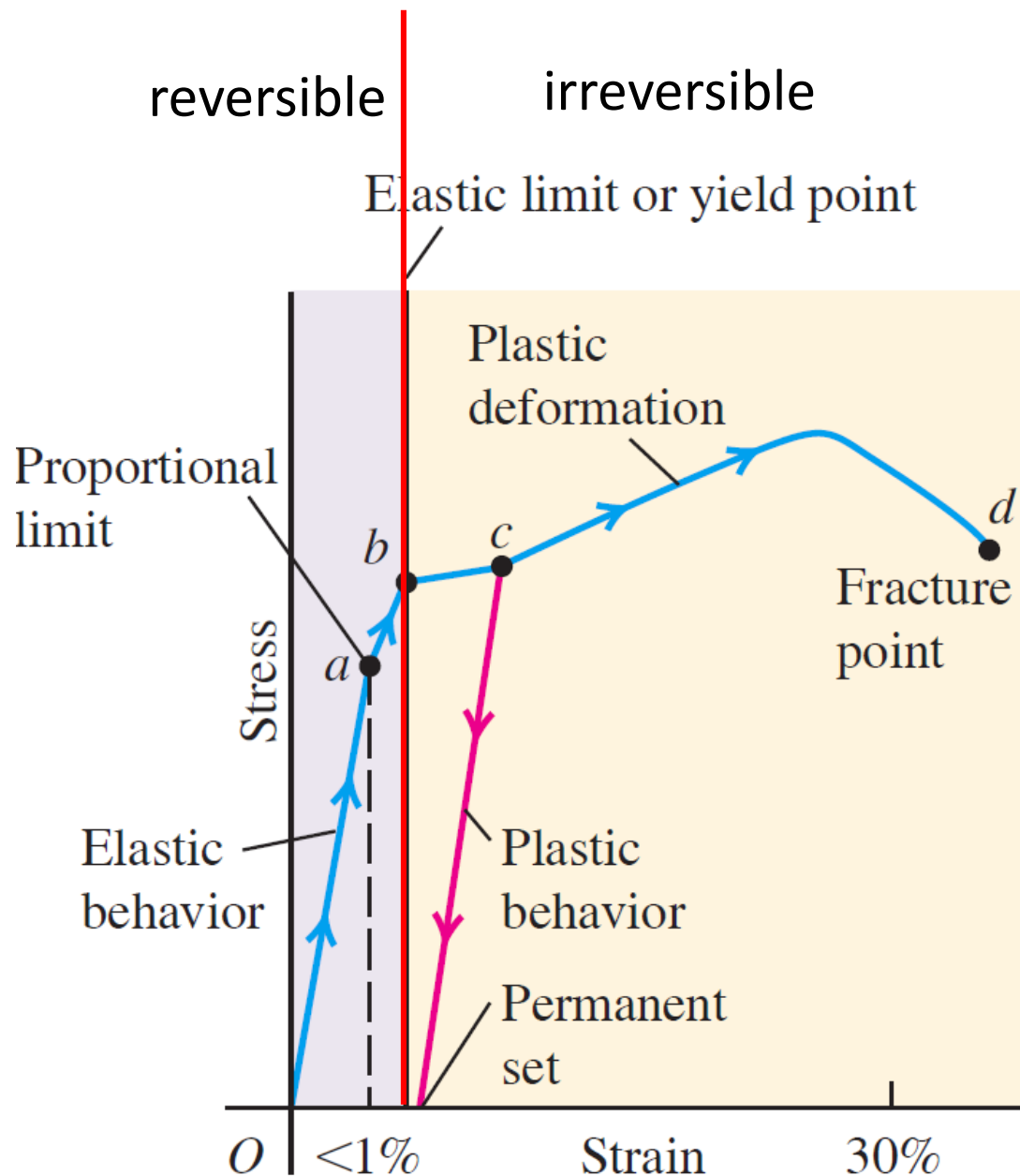
$$S = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F_{\parallel}/A}{x/h} = \frac{F_{\parallel}}{A} \frac{h}{x}$$

Area over which force is exerted

d) 胡克定律的局限性

胡克定律仅适用于弹性形变区，即应变和应力成线性正比

如右图所示：胡克定律仅适用于小应变区间。



脆的 (Brittle)



Fracture point ~ yield point

韧性的 (Ductile)



Fracture point >> yield point

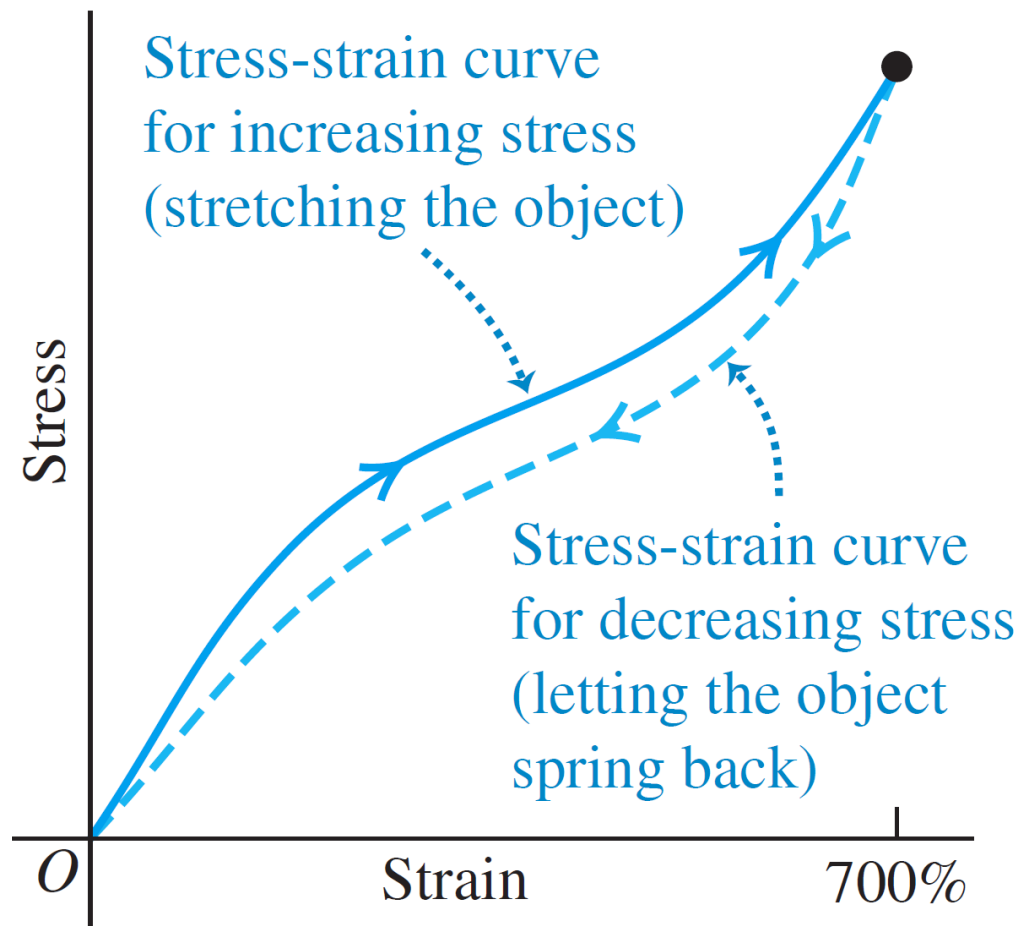
Breaking stress: minimum stress to cause fracture

TABLE 11.3 Approximate Breaking Stresses

Material	Breaking Stress (Pa or N/m²)
Aluminum	2.2×10^8
Brass	4.7×10^8
Glass	10×10^8
Iron	3.0×10^8
Steel	$5-20 \times 10^8$
Tendon (typical)	1×10^8

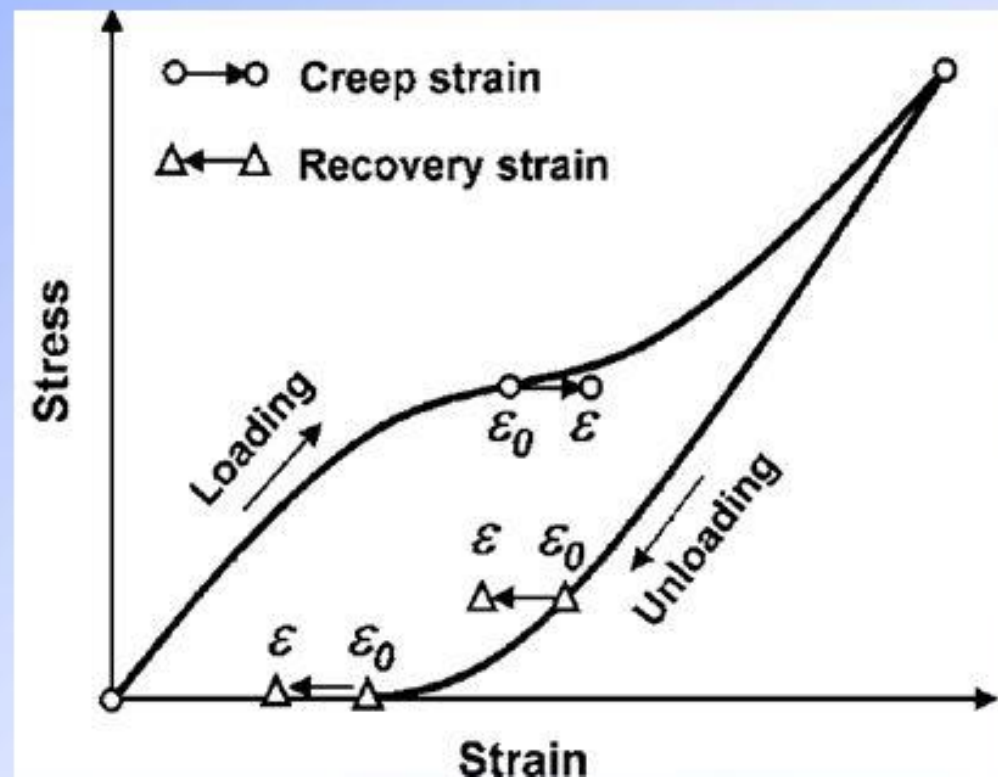
胡克定律不适用的另一情况: 弹性回滞曲线

硫化橡胶
vulcanized rubber



胡克定律不适用的一大类材料: 铁弹材料

- A ferroelastic material exhibits spontaneous strain after application of stress
- Under applied stress a material is deformed
- Ferroelastic materials respond to stress by twinning or phase change



*以下内容拓展内容

平衡

质点 -- $\{x, y, z\}$ 三个运动自由度

质点系 -- $\{x, y, z\} * n$ 个运动自由度

刚体 -- $\{x, y, z\} + \{\alpha, \beta, \gamma\}$ 六个运动自由度 -- 刚体的运动可以分解为质心的平动 + 绕质心的转动

质点的平衡 -- 牛顿第一定律：合力为零

刚体的平衡 -- 合力为零 + 合力矩为零

刚体平衡的拓展

达到平衡 $\rightarrow$ 无进一步形变

形变后的平衡仍然可以用刚体的概念来描述

仅对平衡状态适用



地表对象的平衡：重力场

$$\sum \vec{F} = 0$$

$$Mg = (m_1 + m_2 + m_3 + \dots)g = \sum m_i g$$

$$\sum \vec{\tau} = 0$$

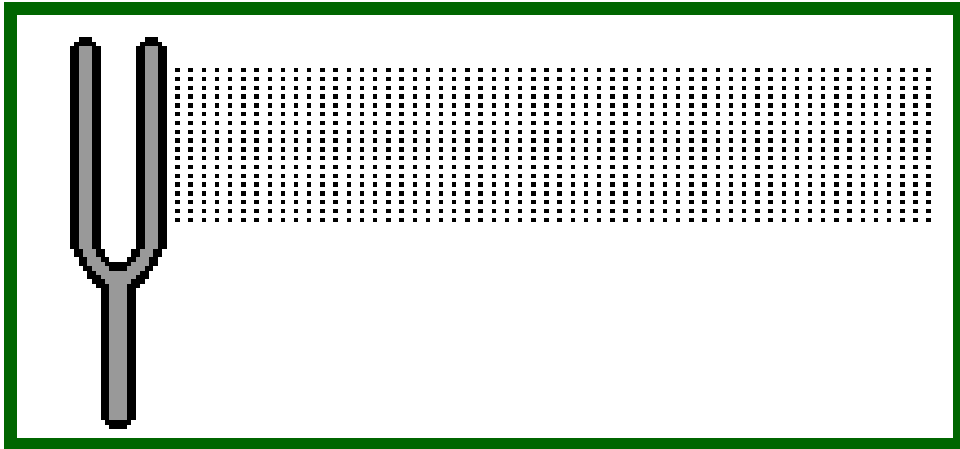
$$\vec{\tau} = \sum_i \vec{\tau}_i = \vec{r}_1 \times m_1 \vec{g} + \vec{r}_2 \times m_2 \vec{g} + \dots = \left(\sum_i m_i \vec{r}_i \right) \times \vec{g} = \vec{r}_{\text{cm}} \times M \vec{g}$$

质心 → 重心：源于重力仅正比于质量，特殊的一种力。



杨氏模量 – 声速 – 声子

我们提到的第一个具体到材料的，宏观，统计的系统参数。



$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

$$v_s = \sqrt{\frac{Y}{\rho}}$$

微观延申，量子化 → 声子